LESSON 2.10

Choose Appropriate Measures

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can choose the best measure of center for a data set based on its shape.

WHY THIS MATTERS The students want to describe the data they collected about the number of years of teaching experience of the teachers in their school. Which measure of center and measure of variation should they use to summarize the data set?

REMEMBER

Choose Appropriate Measures

Choosing Appropriate Measures of Center & Spread The mean and the median both describe the center of a data set, but one fits better depending on the shape. When the data has an outlier (a value far from the rest), the median is usually the better choice.

1. Symmetric Data (No Outliers): Use Mean and MAD.
2. Skewed Data (Has Outliers): Use Median and IQR (outliers distort the mean).
3. Compare distributions using both center (typical value) and spread (variability)

Words: **Mean *(Media)*** · **Median *(Mediana)*** · **Outlier *(Valor atípico)*** · **Skewed *(Sesgado)*** · **Data distribution *(Distribución de datos)***

1 **Circle the letter of the best answer. Show how you know.**

The mean of a data set is 20. A value is 26. What is the absolute deviation?

(A) −6
(B) 6
(C) 26
(D) 46

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

The mean of a data set is 15. One value is 11. What is the absolute deviation of that value from the mean?

(A) −4
(B) 4
(C) 11
(D) 26

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A data set has absolute deviations of 3, 1, 5, 2, 4. What is the MAD?

(A) 1
(B) 3
(C) 5
(D) 15

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Team A has MAD = 2.1 points. Team B has MAD = 6.8 points. Which team is more consistent?

(A) Team B — higher MAD means better performance
(B) Both are equally consistent
(C) Cannot tell from MAD alone
(D) Team A — lower MAD means scores are closer to the mean

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A sports reporter writes: 'The average ticket price for the playoffs is $85.' The actual prices are: $45, $50, $48, $52, $55, $260. The $260 ticket is a courtside seat. What is the mean ticket price?

(A) $50
(B) $51
(C) $85
(D) $260

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

6 **Circle the letter of the best answer. Show how you know.**

A sports reporter writes: 'The average ticket price for the playoffs is $85.' The actual prices are: $45, $50, $48, $52, $55, $260. The $260 ticket is a courtside seat. What is the median ticket price?

(A) $50
(B) $51
(C) $52
(D) $85

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

EXPLAIN YOUR THINKING

What pattern do you notice about when median is the better choice?

SENTENCE STARTER The median is better when the data has ___ because the mean gets pulled toward ___. The mean is better when the data is ___ because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help

LESSON 2.10

Choose Appropriate Measures

Set B · Version B · Second form · core practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each scouting question to the measure that answers it.

_____	What was a typical game when one score is a record-breaking outlier?	A	Median
_____	Which shoe size do most players wear?	B	Mode
_____	On a symmetric, outlier-free set, what is the average score?	C	Mean
_____	How far apart are the highest and lowest sprint times?	D	Range

2 **Write the name of the correct group on each line.**

Sort each measure into whether it describes the CENTER of the data or the SPREAD of the data.

Groups: **Measure of Center** **Measure of Spread**

_____ Mean

_____ Median

_____ Mode

_____ Range

_____ Mean absolute deviation (MAD)

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A sports reporter writes: 'The average ticket price for the playoffs is $85.' The actual prices are: $45, $50, $48, $52, $55, $260. The $260 ticket is a courtside seat. Which measure better describes a 'typical' ticket?

(A) The median, because the $260 outlier pulls the mean far above most prices

(B) The mean, because it uses every value

(C) Neither — you should use the mode

(D) The mean, because $85 is between the extremes

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A scout records points per game: 14, 16, 15, 17, 16, 58. To report a TYPICAL game, which measure of center is best?

(A) Mean, because it uses every game

(B) Range, because it shows the highest game

(C) Median, because 58 is an outlier that pulls the mean up

(D) Mode, because 16 appears twice

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The equipment manager needs to order the MOST COMMON jersey size from this list: M, L, M, S, M, L, M. Which measure answers the question?

(A) Mean, because it averages the sizes

(B) Median, because it finds the middle size

(C) Mode, because it names the value that appears most

(D) Range, because it shows the size gap

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A coach wants to know how SPREAD OUT the team's free-throw makes are: 6, 7, 6, 8, 18. Which measure of spread is most affected by the lone 18?

(A) Range, because it is the highest minus the lowest value

(B) Median, because it is in the middle

(C) Mode, because it is the most common

(D) Mean absolute deviation only, never the range

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What pattern do you notice about when median is the better choice?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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